

Supramolecular Chemistry

Supramolecular Structure Enabled Photo-Responsive Charge Transport in Porphyrin-Based Junctions

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Abstract: Supramolecular electronics offers a valuable opportunity to fabricate functional devices with tunable charge transport (CT) behaviors. Their diverse and non-covalent structures allow for precise modulation of CT with external stimuli. Despite their potential, achieving effective CT modulation in supramolecules through external stimuli remains challenging due to a lack of rigid systems with a predictable response to such stimuli. To address this challenge, we explore the use of porphyrin-based supramolecular structures and their CT responses to light illumination. By incorporating rigid ligands to increase the spatial distance between two porphyrin units, we diminish their interactions, fabricating porphyrin-contained cages that exhibit a higher efficiency of electron-hole pair separation compared to the monomer in solid state. Leveraging this advantage, a more effective photo-responsive CT is achieved for cages in solid-state junctions compared to that of monomers. The current through cage junctions reduces upon illumination. Theoretical calculations indicate that the reduction in conductivity arises from an increased energy offset between the molecule and electrode under light irradiation. Moreover, the photo-responsive CT can be finely tuned by adjusting the metal ion coordinated within the porphyrin and the distance between the cage and the electrode. This research paves a new way for developing supramolecular optoelectronic devices.

Introduction

Molecules are designed and applied as functional components in molecular electronic devices due to their rich structure-related charge transport (CT) behaviors.^[1,2] Supramolecules are emerging as promising candidates for application in functional molecular electronic devices due to their ability to modulate the CT through adaptable intermolecular non-covalent interactions.^[3–12] Moreover, the diverse functionalities of supramolecular architectures also offer great flexibility in CT modulation via external physical stimuli, such as light, electric field, and magnetic field.^[13–16] Compared with other stimuli, light offers unique advantages due to its

high temporal and spatial resolution, non-invasive nature, and ability to operate in solid-state devices.^[17–22] These external modulations endow supramolecular electronics with precise control, enabling the fabrication of functional components like molecular switches.^[23] Despite this promise, the utilization of supramolecular architectures to generate distinctive electrical signals in response to external physical stimuli has, to the best of our knowledge, remained largely unexplored. This limitation likely arises from a lack of stable and rigid supramolecular systems with well-defined, predictable responses to stimuli.

To address this gap, herein we investigate the use of porphyrinic molecules to construct supramolecular structures and to explore their CT responses to light illumination. Compared with photochromic molecules such as azobenzene and spiropyran, porphyrinic derivatives possess rigid cyclic frameworks that provide exceptional stability under illumination, thermal, and chemical conditions, rendering them promising building blocks for supramolecular electronics.^[24,25] Intriguingly, they exhibit strong absorption in the visible range and generate a charge-separated state through electron-hole pair separation upon light irradiation, allowing for facile modulation of their CT.^[26]

In this work, we construct rigid porphyrin cages via supramolecular self-assembly between porphyrinic molecules and a benzophenone structure with a bidentate diplatinum(II) motif (Figure 1a). Our experimental findings indicate that these supramolecular cages markedly improve the efficiency of electron-hole pair separation in porphyrinic moieties, both in solution and solid films. Notably, photo-responsive CT in porphyrin supramolecular cages is observed during

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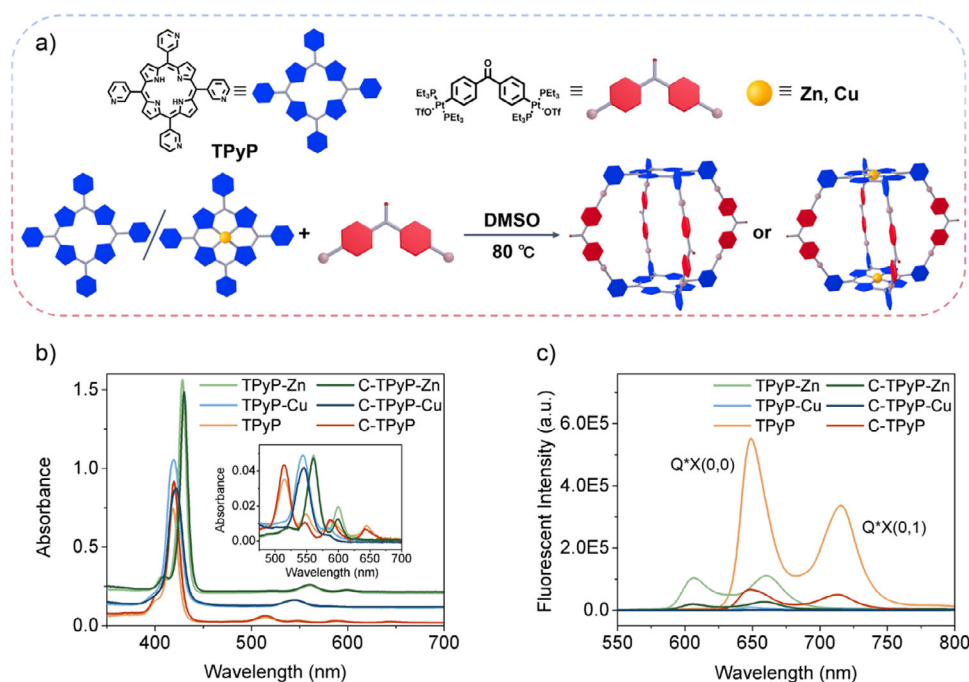


Figure 1. Structures and spectroscopic properties of TPyP cages and monomers. a) Molecular structures of TPyP monomers and cages with the synthetic path. b) UV-vis absorption spectra of TPyP, TPyP-Zn, TPyP-Cu, C-TPyP, C-TPyP-Zn, and C-TPyP-Cu in DMSO. Inset: expanded view of the Q band absorption. c) Emission spectra of TPyP, TPyP-Zn, TPyP-Cu, C-TPyP, C-TPyP-Zn, and C-TPyP-Cu in DMSO under excitation of 420 nm. The concentration of monomers and cages is 3×10^{-6} M and 1.5×10^{-6} M, respectively.

solid-state electrical measurements under light irradiation. There exists a positive correlation between the photo-responsive CT and the efficiency of electron-hole pair separation. The photo-responsive CT can be further explained by the changes in energy gap of the porphyrinic moieties in the charge-separated state relative to the electrode under illumination. Additionally, the photo-responsive CT can be further modulated by altering the type of metal ion chelated into the porphyrinic moieties as well as varying the distance between porphyrinic cages and the gold electrode. This work not only establishes a supramolecular strategy for enhancing electron-hole pair separation efficiency in porphyrins but also elucidates its relationship with photo-responsive CT. These findings offer new insights into the design of optoelectronic molecular devices with tunable functionalities.

Results and Discussion

Porphyrin-contained supramolecular cages (C-TPyP) were synthesized by self-assembling 5,10,15,20-tetrakis(3-pyridyl) porphyrin (TPyP) monomer and diplatinum(II) motif in a 1:2 stoichiometric ratio via coordination between the N of the pyridine group and the Pt of the motif (Figure 1a).^[27] Moreover, TPyP monomers coordinated with Zn^{2+} and Cu^{2+} (TPyP-Zn and TPyP-Cu) were also used to form the supramolecular cages in the same manner (named C-TPyP-Zn and C-TPyP-Cu). These supramolecular cages exhibit an accurate component proportion and high structural stability. In each supramolecular cage, two porphyrin units are positioned as parallel planes on the top and bottom of the cage,

which are separated by four diplatinum(II) motif pillars. The distance between two porphyrin units is about 18.3 Å from the single-crystal X-ray diffraction.^[26]

The ultraviolet-visible (UV-vis) absorption spectra of the three cages were recorded in dimethylsulfoxide (DMSO) (Figure 1b). All the supramolecular cages exhibit distinct absorption bands of the Soret band around 420 nm and Q bands between 475 and 675 nm. The Soret band is attributed to the $\pi-\pi^*$ transition, whereas the Q bands correspond to the absorption transition from the ground state to the first excited singlet state (S_0-S_1). The absorption bands for C-TPyP, as well as C-TPyP-Zn and C-TPyP-Cu, do not display discernible shifts to their corresponding monomers. Compared with TPyP monomer and C-TPyP, the porphyrins with metal ions exhibit slight red shifts for TPyP-Zn and C-TPyP-Zn, while negligible shifts are observed for TPyP-Cu and C-TPyP-Cu in the Soret band. Meanwhile, the four Q band peaks of TPyP monomer and C-TPyP have merged into two due to the increased symmetry of porphyrin upon coordination of metal ions, indicating the coordination of metal ions within the porphyrin structure.

Under the excitation of 420 nm, two distinct emission bands, $Q^*X(0,0)$ and $Q^*X(0,1)$, are observed in supramolecular cages and monomers, which can be assigned as the electronic transition and the associated vibronic transition (Figure 1c).^[28] The fluorescence intensity of C-TPyP is significantly reduced compared to that of the TPyP monomer, while the emission wavelength remains unchanged. With metal ion coordination, the fluorescence intensities of metal-ion-coordinated cages further diminished compared to that of the TPyP monomer. Notably, the fluorescence intensity

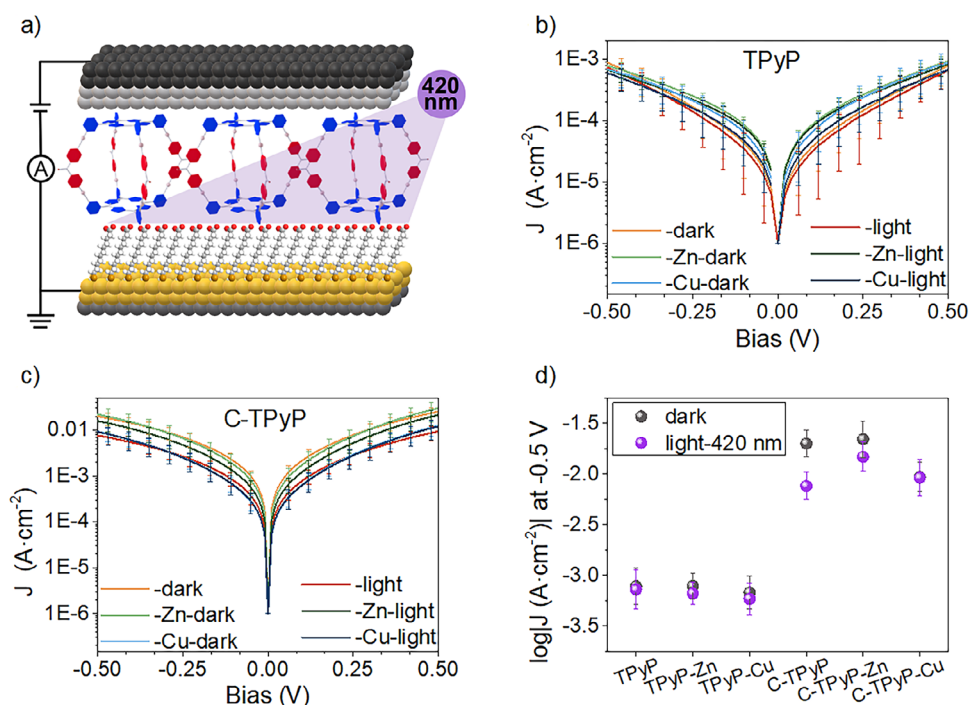


Figure 2. Current–voltage measurements of TPyP cages and monomers under dark or light conditions. a) Schematic of the constructed molecular junction for the current–voltage measurement, wherein the light covers the entire molecular junction. b) Semilog J - V plots for TPyP, TPyP-Zn, and TPyP-Cu under dark or illumination. c) Semilog J - V plots for C-TPyP, C-TPyP-Zn, and C-TPyP-Cu under dark or illumination. d) $\log|J|$ comparison at -0.5 V for TPyP cages and monomers.

of TPyP-Cu monomer was approximately 60 times weaker than that of the TPyP monomer, which can be attributed to the paramagnetic fluorescence quenching between TPyP monomer and Cu^{2+} .^[24] Moreover, the metal-ion-coordinated cages exhibit markedly lower fluorescence than their corresponding monomers, indicating the variations of energy level distribution regulated by TPyP spatial structure and metal-ion coordination.

To fabricate the junctions for electrical measurements, a self-assembled monolayer (SAM) of 11-mercaptoundecanoic acid (MUA, $\text{SC}_{10}\text{COOH}$) was first covalently assembled on a gold substrate as the spacer. A layer of cages (or a film of TPyP monomers) was then assembled (Figure 2a). We conducted stability experiments between the C-TPyP and carboxyl groups in solution. The results confirmed that the cage remains intact in solution without dissociation induced by the carboxyl groups (Figures S3–S6). Based on this, we infer that the structural integrity of the cage is similarly preserved in the solid film. The layers of cages or monomers displayed uniform morphologies, as evidenced by atomic force microscopy (AFM) imaging (Figure S7). The thickness of the cage layers above MUA SAM, as measured by ellipsometry and AFM scratching measurement, was close to the theoretical size of cages, indicating a single layer of TPyP cages densely packed on the MUA SAM with their native structures (Table S1 and Figure S8). For electrical comparison, the thickness of the TPyP monomer film was adjusted to the same as that of the cage SAMs by optimizing the concentration and incubation time. Finally, the $\text{Au-SC}_{10}\text{COOH/TPyP cage/GaO}_x/\text{EGaIn}$ junctions were constructed by loading an EGaIn electrode on

top of the TPyP cage layers for the current density (J)-bias measurements (Figure 2a).

At least 10 junctions and about 15 current density (J)-bias curves for each junction were collected for statistics (Figures 2b and c). The statistical $\log|J|$ values at -0.5 V were extracted for comparison (Figure 2d, Figures S9 and S10). The J value of TPyP cages under dark conditions is more than an order of magnitude higher than that of TPyP monomers with and without coordinated metal ions. When the cage junctions were illuminated with 420 nm light, the J value of C-TPyP decreased by half an order of magnitude, in contrast to the unchanged J value of TPyP monomer. The presence of metal ions coordinated in the center of porphyrin further leads to varying shifts in the J value of the cages. The coordination of Zn^{2+} does not change the current density of the cage, while the coordination of Cu^{2+} slightly decreases the cage current density by around half an order of magnitude. Under illumination, the current of C-TPyP-Zn experiences a slight reduction, while C-TPyP-Cu behaves similarly to TPyP monomer and displays a minimal change in current. During the illumination, the temperature remained nearly constant, making its effect on current changes negligible.

Given that the fluorescence intensity of C-TPyP is markedly weaker than that of TPyP monomer with the same porphyrin unit, it is hypothesized that the photo-responsive CT of C-TPyP might be related to the charge-separated state of porphyrin. Generally, a weaker fluorescence emission intensity correlates with a higher efficiency of electron-hole pair separation.^[29,30] The weaker fluorescence emission intensity of C-TPyP, compared to TPyP monomer, implies

that the effective separation of porphyrin units in C-TPyP may enhance the efficiency of electron-hole pair separation and lead to the current changes under illumination. Moreover, the fluorescence intensities of C-TPyP-Zn and C-TPyP-Cu are also weaker than those of their corresponding TPyP-Zn and TPyP-Cu monomers, indicating more efficient electron-hole pair separation in these cages compared to their monomers, consistent with the enhanced separation behaviors observed in C-TPyP relative to the TPyP monomer. However, the TPyP-Zn and TPyP-Cu monomers exhibit lower fluorescence intensities compared to the uncoordinated TPyP. The low fluorescence can be attributed to both electron-hole pair separation and coordinated metal ions. The decrease in fluorescence emission intensity alone is not suitable to evaluate the electron-hole pair separation efficiency among these three monomers or the three cages, respectively.

To further assess the efficiency of electron-hole pair separation in the solid-state single layer of the supramolecular cages, the contact potential differences (CPD) of the TPyP cages and monomers were measured by Kelvin probe force microscopy (KPFM) under a N_2 atmosphere. The surface photovoltage (SPV) was then calculated from $CPD_{dark} - CPD_{light}$ (Figure 3a, Figures S11 and S12). All the TPyP supramolecular cages and monomers exhibit negative SPV. The absolute SPV value (-SPV) of C-TPyP is much larger than TPyP monomer. When a metal ion coordinates to the cage, the -SPV values decrease in the sequence of C-TPyP > C-TPyP-Zn > C-TPyP-Cu. The -SPV of C-TPyP-Cu is close to the -SPV of TPyP monomers with and without metal ion coordinators. The -SPV shows a similar trend with the photo-responsive current density defined as $\log(J_{dark} - J_{light}/J_{light})$.

SPV reflects the photo-induced dipole generated by electron-hole pair separation under illumination in solid films. The polarity of SPV describes the direction of the photo-induced carrier redistribution on the surface, and a larger absolute value indicates a higher efficiency of electron-hole pair separation.^[31] In the KPFM measurements, a smaller CPD corresponds to an increased V_{sample} , indicating a higher work function (W_F) of the sample. The negative SPV values indicate that a reduction of W_F for TPyP supramolecular cages and monomer under illumination, suggesting a tendency for excited electrons to migrate from the porphyrin units to the gold substrate in the films (Figure 3b). The assembled C-TPyP showed a larger negative SPV value of approximately 0.25 V compared with the TPyP monomer film. The porphyrin units in C-TPyP are effectively separated by a distance of 18.3 Å through the rigid diplatinum(II) motifs, which is significantly larger than the approximately 5.5 Å distance observed in the packed TPyP structures in the monomer film.^[32] The cage structure weakens the intermolecular stacking, leading to a higher efficiency of electron-hole pair separation under illumination. Consequently, compared to TPyP monomer, C-TPyP demonstrates enhanced efficiency of electron-hole pair separation in solid films under illumination. Moreover, the SPVs of C-TPyP-Zn and C-TPyP-Cu are significantly lower than that of C-TPyP, suggesting that the efficiency of electron-hole pair separation of the cages in the solid-state is also influenced by the metal ion chelated to the porphyrin

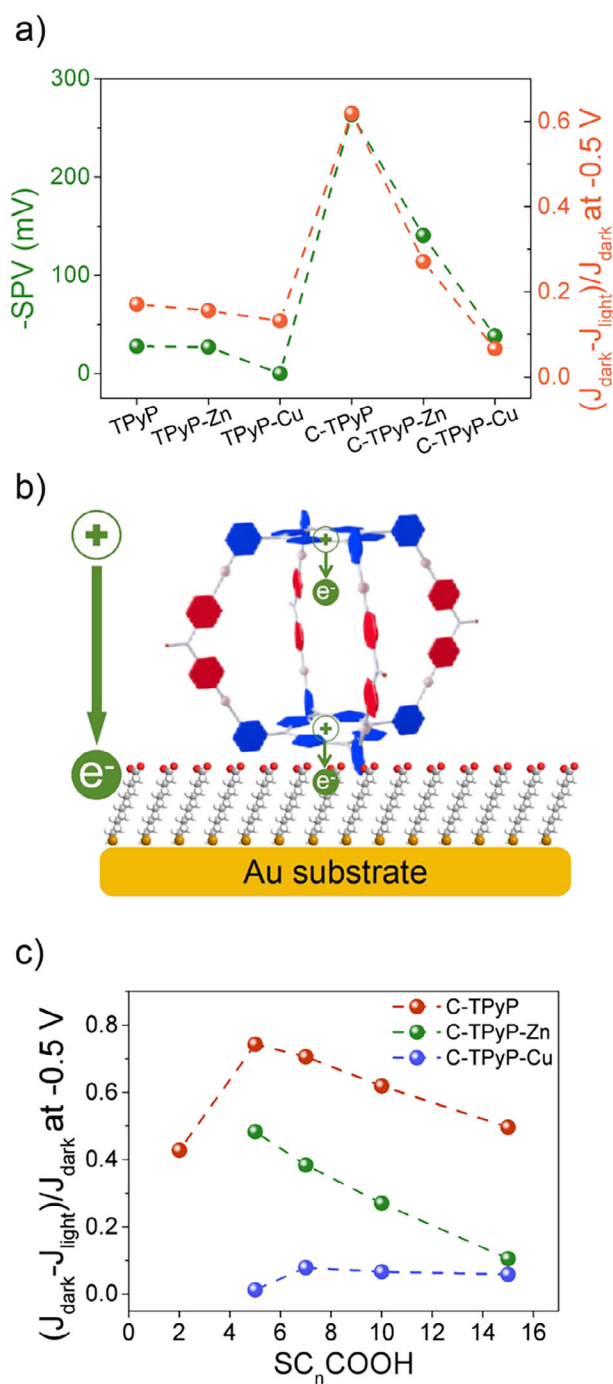


Figure 3. SPV and photo-responsive CT of TPyP cages and monomer. a) The SPV under illumination measured by KPFM and the photo-responsive CT for TPyP monomer and cage layers with and without metal ion coordination. b) Schematic structure of photo-induced dipole in the film. The direction of polarization is dictated by the negative values of SPV. c) Plots of photo-responsive CT of C-TPyP, C-TPyP-Zn, and C-TPyP-Cu with different length of spacers.

units. The similar trends between the negative SPV and photo-responsive current density of the TPyP supramolecular cages and monomers suggest that the photo-responsive CT would be determined by the efficiency of electron-hole pair separation in the supramolecular cages.

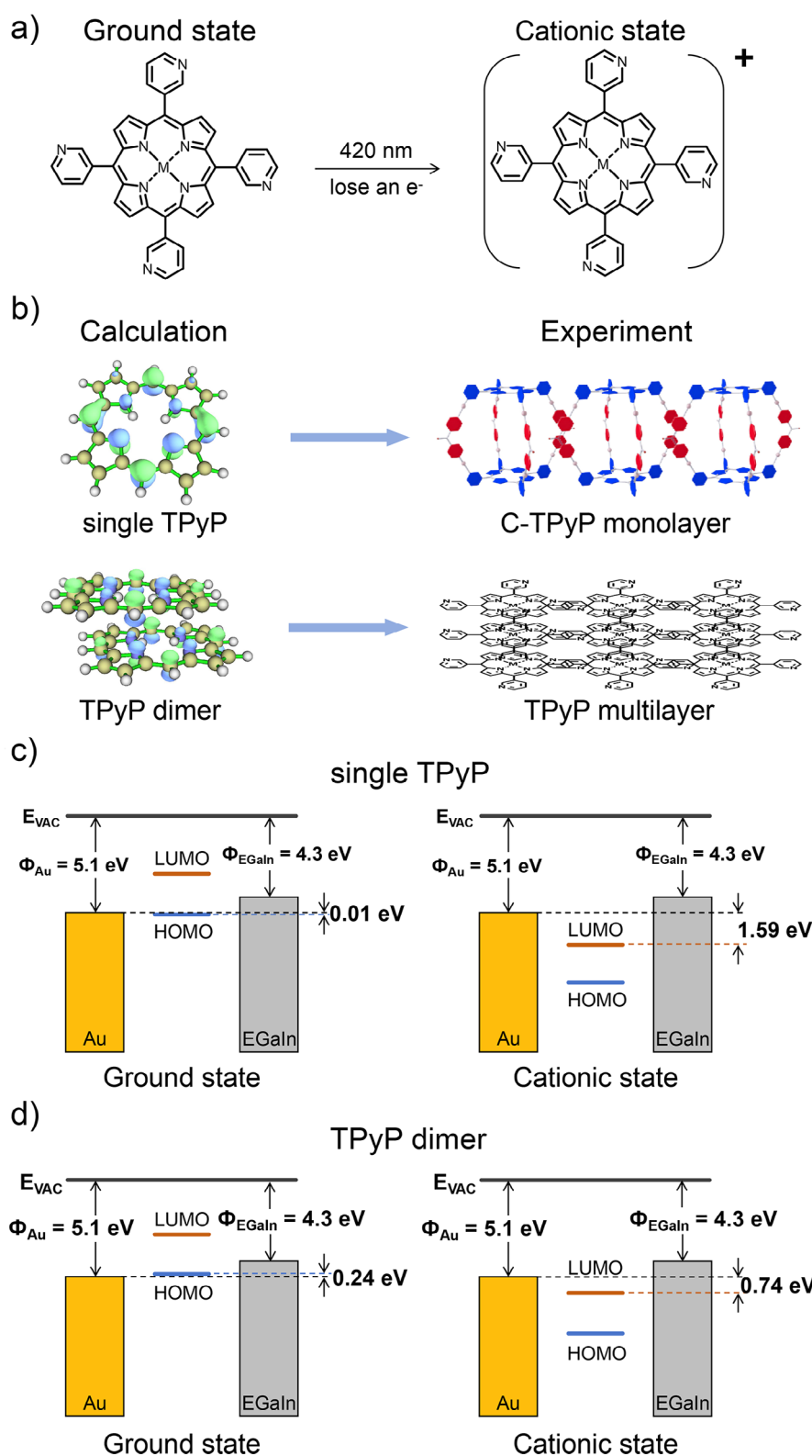


Figure 4. Molecular energy level shifts by illumination. a) Porphyrin unit loses an electron under illumination and changes to a cationic state from the ground state. b) In the DFT calculations, the single TPyP represents the C-TPyP monolayer observed in the experiment, while the TPyP dimer corresponds to the TPyP multilayer in the experiment. c,d) Energy level diagrams of single TPyP and TPyP dimer connecting between Au and EGaIn electrodes in an open circuit.

The distance between the supramolecular cages and electrodes would also influence the photo-responsive CT in solid-state, as the excited electrons migrate between the porphyrin units and electrodes (Figure 3b). To explore this, we selected SC_nCOOH molecules with varying chain lengths ($n = 2, 5, 7, 10, 15$) as spacers to construct the supramolecular cage junctions and compared the CT via junctions under dark and light conditions (Figures S13–S15). For SC_nCOOH with n ranging from 5 to 15, the photo-responsive CT of C-TPyP and C-TPyP-Zn decreases as the spacer length increases (Figure 3c). The increased distance between the porphyrin units of the cage and the Au electrode inhibits the electron-hole pair separation, thereby reducing the photo-responsive CT. On the other side, when the distance is significantly reduced, as in the case of the SC_2COOH spacer in C-TPyP junction, the photo-responsive CT drops dramatically. This effect is likely due to the over-strong coupling between the porphyrin unit of the cage and the electrode, which suppresses the molecular properties,^[33,34] and may lead to disintegration of the TPyP supramolecular cages. In contrast to C-TPyP and C-TPyP-Zn, C-TPyP-Cu junction exhibits weak photo-responsive CT regardless of spacer length. This is likely attributed to the inherently low efficiency of electron-hole pair separation in C-TPyP-Cu.

As the electron-hole pair separation occurs in the C-TPyP solid-state film under illumination, the energy level of C-TPyP may shift, thereby attenuating the CT through the C-TPyP junction. In the charge-separated state under illumination, the excited state of porphyrin units is considered to be cationic, with one electron lost (Figure 4a). The highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) energy levels of C-TPyP as well as its cationic state were evaluated through the density functional theory (DFT) calculations (Table S2). The calculated HOMO–LUMO gap for the ground state of C-TPyP is 0.03 eV, much smaller than the optical HOMO–LUMO gap of 1.85 eV measured from absorption spectroscopy (Figure S16). There is a substantial discrepancy between the calculation and the experiment. From the absorption and fluorescence spectra, the energy levels of C-TPyP are mainly determined by the porphyrin units. The main difference between C-TPyP and TPyP monomer for photo-responsive CT is the distance between the porphyrin units. Therefore, a single TPyP was set as the calculation model to simulate the situation of C-TPyP with spatially separated two porphyrin units. A TPyP dimer with an optimized TPyP distance of 3.4 Å was set as the model to simulate the case of TPyP monomers in a multilayer packing structure (Figure 4b). The calculated HOMO–LUMO gap for the single TPyP is 2.10 eV, close to the optical HOMO–LUMO gap of 1.85 eV for C-TPyP. The 0.25 eV difference may arise from the various factors in optical measurements, such as electron-hole recombination and electron excitation, leading to a reduction in the HOMO–LUMO gap.^[35]

From the DFT calculations, the energy levels of ground and cationic states exhibit significant differences for both single TPyP and TPyP dimer, leading to noticeable shifts relative to the Fermi level of the electrodes (Figure 4c,d, Table S2). For the single TPyP model that simulates C-TPyP, the

HOMO of single TPyP in the ground state is close to the Au electrode, with an energy barrier of 0.01 eV, while in the cationic state, the LUMO is near the nearest Au electrode, with an energy barrier of 1.59 eV. Therefore, the CT through the C-TPyP junction is attenuated under illumination. For the TPyP dimer model that mimics the TPyP monomer, the CT of TPyP dimer also shifts from HOMO to LUMO dominance as the dimer transitions from the ground state to the cationic state under illumination. In this process, the energy barrier between the nearest energy level of TPyP dimer and the closest Au electrode increases from 0.24 to 0.74 eV. As a result, a reduction in CT is expected through the TPyP monomer junction under illumination. However, due to the low efficiency of electron-hole pair separation in TPyP monomer, fewer cationic states are generated under illumination, ultimately resulting in minimal changes in CT.

Conclusion

In this study, we have successfully fabricated a series of molecular junctions based on supramolecular cages composed of TPyP units to investigate their photo-responsive CT behaviors. A pronounced photo-responsive current decrease was observed on C-TPyP in solid-state under 420 nm light irradiation. Both the metal ions coordinated in porphyrin units and the distance between the TPyP cage and the electrode can finely modulate the photo-responsive CT. Demonstrated by the SPV results, this photo-responsive CT behavior exhibits a positive correlation with the efficiency of electron-hole pair separation within the TPyP cages. The introduction of rigid skeletons increases the distance between two porphyrin units, thereby diminishing their mutual interactions and consequently enhancing the efficiency of the electron-hole pair separation in solid states. From the theoretical calculations, the energy offset increases between the excited TPyP cages and the electrodes. The efficient electron-hole pair separation, combined with the increased energy offset in excited TPyP cages, collectively contributes to the illumination-induced conductivity reduction in the solid-state. This research not only advances our understanding of the photo-responsive CT mechanism in porphyrin-based molecular junctions, but also highlights the potential applications of metal-coordinated supramolecular cages for applications in optoelectronic devices within molecular electronics.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: Molecular electronics • Photo-response • Porphyrins • Self-assembly • Supramolecules

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